MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 7322

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen, ceramic-metal

See figure 1 and note 1

Mounting position: Any

Weight: 2 pounds (907.2 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Eres	Ebb	Ecc	epy	epx	egy	egx	Ib
Unit:	V	Vac	Vdc	Vdc	kv	kv	v	v	A
Maximum:	6.8	6.8	---	-200	25	25	1,500	400	2.0
Minimum:	5.8	5.8	1,500	0	--- (Note 2)	5% epy (Note 3)	500 (Note 4)	--- (Note 5)	---
TEST CONDITIONS:	6.3	6.3	---	0	25	---	450	---	---

ABSOLUTE RATINGS:

Parameter:	Ip	ib	dik/dt	Pb	prr	tj	tk	TA	Cooling
Unit:	Aac	a	a/µs	---	---	µs	sec	°C	---
Maximum:	36	1,000	5,000	20×10^9	---	0.005	---	+125	Note 6
Minimum:	---	---	---	---	---	---	---	-55	---
TEST CONDITIONS:	---	---	---	---	500	---	600	Ambient	---

GENERAL:

Qualification - Required

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3267	Instantaneous starting	7,8	Ef = Eres = 6.8 Vac; epy = 18 kv (min); 1st test after holding period	0.65	II	---	---	---	---
3246	Operation (1)	7,9	Ef = Eres = 5.8 Vac; epy = 27.5 kv; 2nd test after holding period	0.65	II	---	---	---	---
3247	DC anode voltage for conduction	-	Ef = Eres = 5.8 Vac	0.65	II	Ebb	---	1,000	Vdc
3246	Operation (2A)	-	Operation (2), except Ef = Eres = 6.8 Vac	0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
3241	Heater current (cathode)	-		---	---	If	11.0	22.0	Aac
3241	Heater current (reservoir)	-		---	---	Ires	3.0	6.0	Aac
3246	Operation (1A)	-	Operation (1), except Ef = Eres = 6.8 Vac; t = 20 minutes	---	---	---	---	---	---
3256	Anode delay time	-	Operation (1); t = 120	---	---	tad	---	0.5	µs
3256	Anode delay time drift	11	Anode delay time	---	---	Δtad	---	0.15	µs
3261	Time jitter	12	Operation (1); Ef = Eres = 5.8 Vac	---	---	tj	---	0.005	µs
3246	Operation (2)	13	Ef = Eres = 5.8 Vac; epy = 20 kv; prr = 1,500	---	---	---	---	---	---
3251	Pulse emission	-	ik = 1,500a; tp = 5.0 µs ±10%; prr = 60 ±10%; tr = 0.5 µs (max); starting time of current pulse = 2.5 µs	---	---	egk	---	250	v
	Quality conformance inspection, part 3								
---	Life test (1)	7,14, 15	Group C; t = 500 hours	---	---	---	---	---	---
---	Life test (2)	13,14, 15	Group C; t = 500 hours	---	---	---	---	---	---
---	Life-test (1) and (2) end points:								
3246	Operation (1) and (1A) except note 13	-		---	---	egy	---	450	v
3247	DC anode voltage for conduction	-		---	---	Ebb	---	1,250	Vdc
3261	Time jitter	-		---	---	tj	---	0.005	µs
3246	Operation at elevated ambient temperature	7,16 17	TA = 125°C	---	---	t	5	---	hrs

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3 -Continued								
3246	Operation (4)	17	epy = 13.5 kv (min); ib = 100a (min); Ib = 2.0 A (min); Pb = 0.4 x 10^9 (min); tp = 60 µs ±10%; tr = 2.5 µs (max); grid pulse same as note 7	---	---	---	---	---	---
3246	Operation (5)	10,17		---	---	---	---	---	---
1031	Sweep-frequency vibration	17	No voltages applied	---	---	---	---	---	---
1041	Shock	17	Accel = 250 G; no voltages applied	---	---	---	---	---	---
---	Shock and vibration end points:								
3247	DC anode voltage for conduction	-		---	---	Ebb	---	1,250	Vdc
3246	Operation (1)	-		---	---	egy	---	450	v
3261	Time jitter	-		---	---	tj	---	0.005	µs
---	Life test (3)	14,17	Operation (4); t = 500 hours	---	---	---	---	---	---
---	Life test (3) end points:								
3246	Operation (4)	-		---	---	egy	---	450	v
3246	DC anode voltage for conduction	-		---	---	Ebb	---	1,250	Vdc

NOTES:

1. An MT-4 mounting assembly (or equivalent) shall be provided as an integral part of each tube. (See figures 2 and 3 for reference only.)
2. Instantaneous starting is permissible. The maximum permissible instantaneously applied epy is 18.0 kv and shall be attained in not less than 0.04 second. The epy may then be raised to full rating.
3. In pulsed operation, the peak inverse voltage exclusive of a spike of 0.05 µs maximum duration, shall not exceed 5.0 kv during the first 25 µs following the anode pulse.
4. The driver pulse, measured at tube socket with thyratron grid disconnected: tr = 0.35 µs maximum, grid pulse duration = 2.0 µs minimum. Impedance of drive circuit 50 to 400 ohms.
5. A maximum backswing voltage of 650 volts on the grid of the thyratron is permissible immediately after the trigger pulse.
6. There shall be no forced-air cooling which will reduce T (anode) or TE below 150°C. At average current levels exceeding 1.5 amperes when the Pb factor exceeds 10 x 10^9, cooling of the tube is necessary by means of an anode radiator, forced air, or both. Radiator design for operation at Ib = 2.0 Adc and Pb = 20 x 10^9 shall be such that the center of the upper radiator surface shall not exceed 200°C under operation. A cooling airblast of 10 cfm may be directed into the anode cup.
7. The circuit constants shall be chosen under resonant charging conditions so that: epy = 25 kv; ib = 1,000a minimum; dik/dt = 5,000 a/µs minimum; tp = 2.5 ±0.25 µs; prr = 500 minimum. Grid pulse shall have the following characteristics: tr = 0.35 µs minimum, tp = 2.0 µs maximum, Zs = 50 to 400 ohms.
8. The tube shall operate satisfactorily on pushbutton starting within three attempts when the epy is applied to the tube under test in such a manner as to rise from 0 to 18 kv minimum within 0.03 second. (The filter in the rectifier shall be designed so that the epy reaches at least 9 kv within 0.015 second.) Any tube failing to start within three attempts will be considered a failure.
9. The tube shall operate continuously for 30 minutes without evidence of arc-back, with not more than three kick-outs during the first minute of operation.
10. Reduce the ambient temperature to -55°C in not less than 30 minutes. After 15 minutes soaking, tubes shall perform to operation (1) conditions, except Ef = Eres = 6.3 V.

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NOTES: -Continued

11. This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 minutes, 4 minutes, and 30 minutes of the operation (1) test. The Δtad measurement is the numerical difference between the 2-minute and 4-minute, or 2-minute and 30-minute tad readings, whichever is greater.

12. The tube shall be tested by applying a peak forward anode voltage not to exceed that specified in the test conditions for the time jitter test immediately after the tk. The tj shall be not greater than the amount specified after 120 seconds of operation.

13. The anode circuit constants shall be so chosen that the epy = 20 kv; ib = 667a minimum; dik/dt = 5,000 a/μs minimum; tp = 1.0 μs ±10 percent; prr = 1,500 minimum; grid pulse same as note 7. Tube shall operate for 20 minutes without continuous conduction.

14. During every 100 +24 hours of life-test operation, the life test shall be shut off for a minimum of 60 minutes, and then checked for life-test end points.

15. Where production is less than 50 tubes per month, life test (1) will be performed alternately with life test (2) on a monthly basis. Sample size under these conditions shall total one tube per month.

16. Tube must operate satisfactorily at the specified ambient temperature at operation (1) condition for a total of 5 consecutive hours with no more than three kickouts and with no evidence of detrimental anode heating. The tube shall be started with Ef = Eres = 6.8 Vac and operate at this value for 4 hours. At the start of the fifth hour, and while the tube is still operating, the filament voltage shall be lowered to Ef = Eres = 5.8 Vac and remain there for the final 1 hour of operation.

(F) 17. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

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Air Force - 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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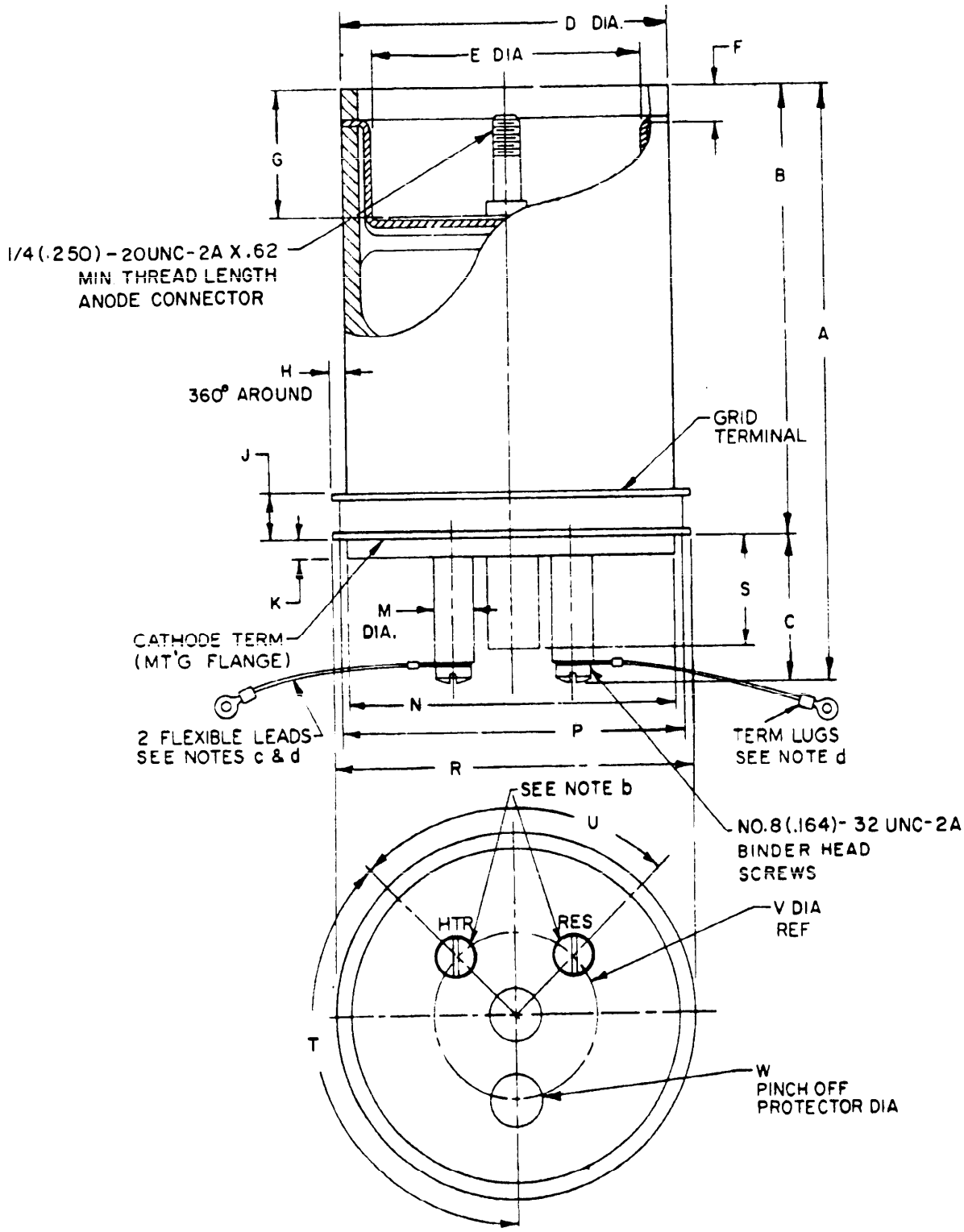


FIGURE 1. Outline drawing of electron tube type 7322.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
	Quality conformance inspection, part 1 (see note a)	
K	.100 (2.54)	.180 (4.57)
N	2.985 (75.82)	3.015 (76.58)
	Quality conformance inspection, part 2	
A		5.895 (149.73)
B	3.945 (100.20)	4.340 (110.24)
C		1.555 (39.50)
D	2.985 (75.82)	3.015 (76.58)
E	2.410 (61.21)	2.530 (64.26)
F		.313 (7.95)
G	.980 (24.89)	1.355 (34.42)
H	.093 (2.36)	
J	.520 (13.21)	.570 (14.48)
M		.375 (9.53)
P	3.235 (82.17)	3.265 (82.93)
R	3.245 (82.42)	3.305 (83.95)
S		1.081 (27.46)
W		.469 (11.91)
	Reference dimensions	
T	135°	
U	90°	
V	1.500 (38.10)	

NOTES:
a. The AQL for dimensions listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
b. Reservoir and heater terminals shall be identified and marked on tube base as specified herein.
c. Insulated leads to be on 10.0 (254. mm) ±1.0 (25.4 mm) centers.
d. Lead identification terminal lugs and insulation sleeving (in accordance with MIL-I-3190) shall be as follows: See note e for inspection procedures.

Lead	Color	Sleeving	Terminal lugs clearance hole
H - Heater	Yellow	H-A-I or H-B-I	1/4 stud
R - Reservoir	Red	H-A-I or H-B-I	#10 stud

(F) e. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 7322 - Continued.

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INCHES	MM
.005	.13
.012	.30
.015	.38
.219	5.56
.428	10.87
.856	21.74
1.931	49.05
3.062	77.77
3.861	98.07

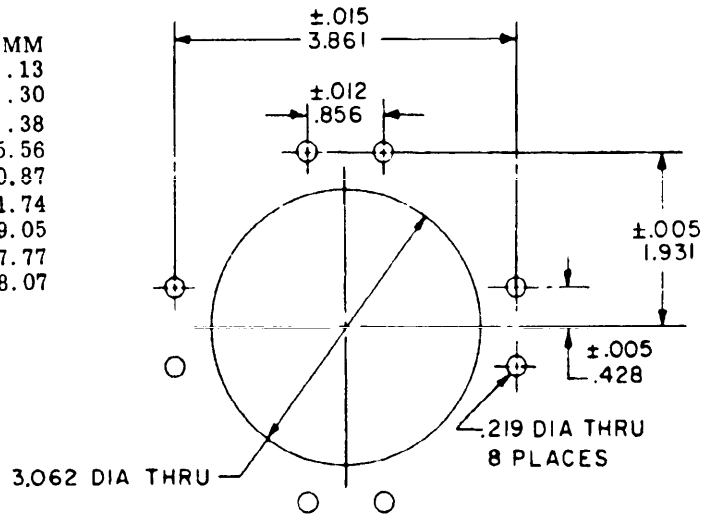


FIGURE 2. Mounting hole dimensions.

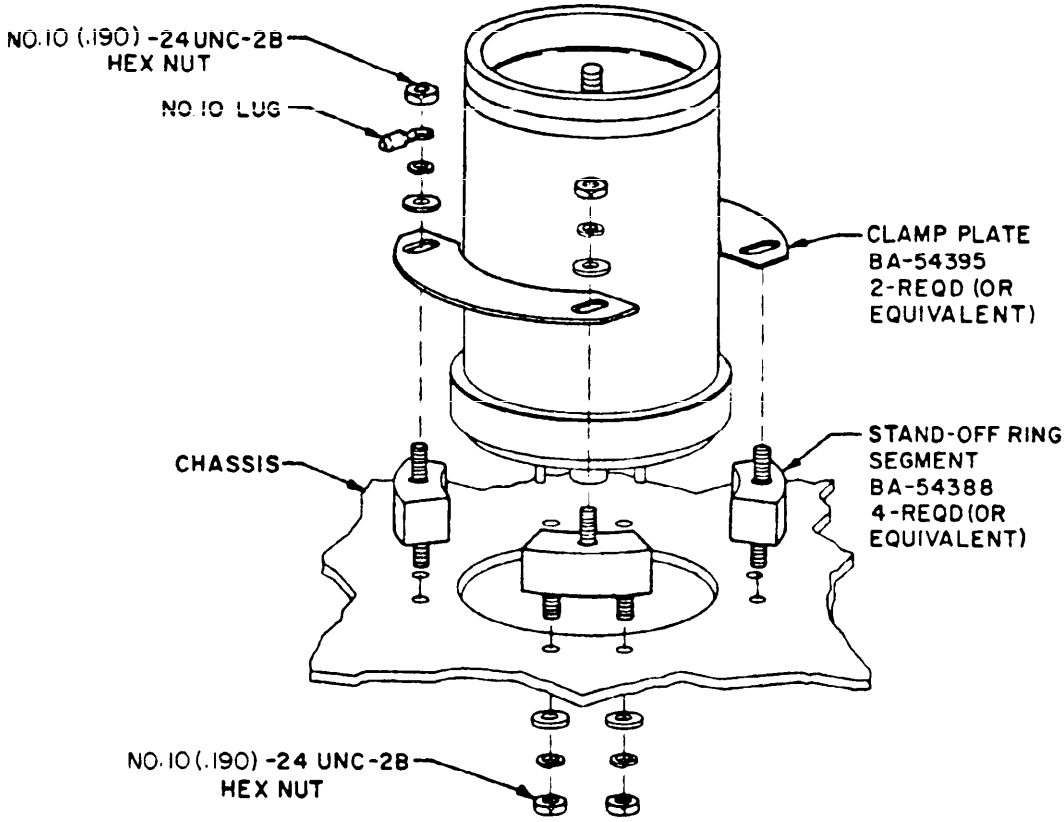


FIGURE 3. MT-4 tube mounting assembly.

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